Pizza queue

- A Pizza stand receives on average 3 orders per 10-minute interval at a sports event.
- As the owner, you are interested in understanding certain probabilities and would like to make use of Queuing Theory.
- 1) What assumption must you make regarding the number of orders received within any time window?

Pizza queue

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- As the owner, you are interested in understanding certain probabilities and would like to make use of Queuing Theory.
- 1) What assumption must you make regarding the number of orders received within any time window? **The number of orders follows a Poison distribution.**

Pizza queue

- Provided the inter arrival times follows an exponential distribution, attempt the following questions

  *remember $P(N = n) = {}^{(e^{-\lambda})\lambda^n}/_{n!}$
- What is the probability that:
  - 0 pizza will be ordered in 10 minutes?
  - 1 pizza will be ordered in 10 minutes?
  - 2 pizza will be ordered in 10 minutes?
  - 3 pizza will be ordered in 10 minutes?
  - 6 pizza will be ordered in 20 minutes?
  - 9 pizza will be ordered in 30 minutes?

Pizza queue

- Provided the inter arrival times follows an exponential distribution, attempt the following questions

  *remember $P(N = n) = {}^{(e^{-\lambda})\lambda^n}\!/_{n!}$

- Note you can 'customise' your λ or wite out in terms of factor of "t": $\boldsymbol{P(N_t = n) = {}^{e^{-\lambda t}(\lambda t)^n}\!/_{n!}}$

Pizza queue

- What is the probability that:
  - 0 pizza will be ordered in 10 minutes? $\boldsymbol{P(N_t = 0) = {}^{e^{-3}(3)^0}\!/_{0!} = 0.0498}$
  - 1 pizza will be ordered in 10 minutes? $\boldsymbol{P(N_t = 1) = {}^{e^{-3}(3)^1}\!/_{1!} = 0.1494}$
  - 2 pizza will be ordered in 10 minutes? $\boldsymbol{P(N_t = 2) = {}^{e^{-3}(3)^2}\!/_{2!} = 0.2240}$
  - 3 pizza will be ordered in 10 minutes? $\boldsymbol{P(N_t = 3) = {}^{e^{-3}(3)^3}\!/_{3!} = 0.2240}$
  - 6 pizza will be ordered in 20 minutes?
  - 9 pizza will be ordered in 30 minutes?

Pizza queue

- What is the probability that:
  - 6 pizza will be ordered in 20 minutes?
  - 9 pizza will be ordered in 30 minutes?

*Note you can either 'customise' your λ and determine a new one for each question or wite out it out in terms of factor of "t": $\boldsymbol{P(N_t = n) = {e^{-\lambda t}(\lambda t)^n}/{n!}}$

Pizza queue

- What is the probability that:
  - 0 pizza will be ordered in 10 minutes? $\boldsymbol{P(N_t = 0) = {}^{e^{-3}(3)^0}/_{0!} = 0.0498}$
  - 1 pizza will be ordered in 10 minutes? $\boldsymbol{P(N_t = 1) = {}^{e^{-3}(3)^1}/_{1!} = 0.1494}$
  - 2 pizza will be ordered in 10 minutes? $\boldsymbol{P(N_t = 2) = {}^{e^{-3}(3)^2}/_{2!} = 0.2240}$
  - 3 pizza will be ordered in 10 minutes? $\boldsymbol{P(N_t = 3) = {}^{e^{-3}(3)^3}/_{3!} = 0.2240}$
  - 6 pizza will be ordered in 20 minutes? $\boldsymbol{P(N_t = 6) = {}^{e^{-3\times 2}(3\times 2)^6}/_{6!} = 0.1606}$
  - 9 pizza will be ordered in 30 minutes? $\boldsymbol{P(N_t = 9) = {}^{e^{-3\times 3}(3\times 3)^9}/_{9!} = 0.1318}$

Pizza queue

- What is the probability that:
  - Up to three people will arrive within 10 minutes?
  - Between 4 and 6 people will arrive in the next 20 minutes?

- What is the probability that:
  - Between 8 and 15 minutes will pass between any two orders?

*Remember inter arrival times follow an exponential distribution but the number of arrivals in an interval is a Poisson process.

Pizza queue

- What is the probability that:
  - Up to three people will arrive within 10 minutes? For $\lambda = 3$:

$$\sum_{i=0..3} \boldsymbol{P(N_t = i) = 0.6472}$$

  - Between 4 and 6 people will arrive in the next 20 minutes? For $\lambda = 6$:

$$\sum_{i=4,5,6} \boldsymbol{P(N_t = i) = 0.4551}$$

  - What is the probability that:
  - Between 8 and 15 minutes will pass between any two orders? For $\lambda = \frac{3}{10}$:

$$\boldsymbol{\int_{t=8}^{t=15} \lambda e^{-\lambda t} = 0.0796}$$

Tiger and Wheels – service times

- A Tiger and Wheels station can perform wheel alignment for your motor vehicle at an average rate of 20 minutes. A customer, who requires wheel alignment to be performed at the station, typically arrives every 30 minutes during normal business hours. Now calculate the following:
  - λ and μ
  - The average number of customers present in the system, call it "L".
  - The average number of customers in the queue, call it "L_q".
  - The average time a customer spends in the system, call it "W".
  - The average time a customer spends waiting in the queue, call it "W_q".
  - The percentage of time that a station is busy.
  - The probability that 0 cars are in the system, call it "P_0".

Tiger and Wheels – service times

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  - λ and μ
    - $\lambda = 2$ cars arriving per hour.
    - $\mu = 3$ cars serviced per hour.

Tiger and Wheels – service times

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  - The average number of customers present in the system, call it "L". (Hint, use steady state service)

Tiger and Wheels – service times (M/M/1/GD/∞/ ∞)

- The average number of customers present in the system, call it "L"

$\boldsymbol{L = \frac{\rho}{(1-\rho)} =}$ **2 cars on average, where** $\boldsymbol{\rho = \frac{\lambda}{\mu} = \frac{2}{3}}$

- The average number of customers in the queue, call it "L_q".

$\boldsymbol{L_q = \frac{\rho^2}{(1-\rho)} = 4/3}$ **cars waiting in line on average.**

- The average time a customer spends in the system, call it "W".

$\boldsymbol{W = \frac{L}{\lambda} = \frac{1}{(1-\rho)\mu} = 1}$ **hour the average car spends in the system.**

- The average time a customer spends waiting in the queue, call it "W_q".

$\boldsymbol{W_q = \frac{\rho}{\mu(1-\rho)} = 2/3}$ **hours average waiting time per car.**

Tiger and Wheels – service times (M/M/1/GD/∞/ ∞)

- The percentage of time that a station is busy.

$$\boldsymbol{\rho = \frac{\lambda}{\mu} = 2/3}$$

- The probability that 0 cars are in the system, call it “P_0”.

$$\boldsymbol{P_0 = 1 - \frac{\lambda}{\mu} = 1/3}$$

Tiger and Wheels – service times

- Management wants to investigate the financial feasibility of opening a second station with a second technician. This station will be equal in terms of quality of work, as well as the time it takes to perform it. Now calculate the following:
  - λ and μ
  - The average number of customers present in the system.
  - The average number of customers in the queue.
  - The average time a customer spends in the system.
  - The average time a customer spends waiting in the queue.
  - The percentage of time that a station is busy.
  - The probability that 0 cars are in the system.

Tiger and Wheels – service times (M/M/2/GD/∞/ ∞)

- $\lambda = 2$ [cars/hour] and $\mu = 3$ [cars/hour] with $s = 2$ parallel services.

Tiger and Wheels – service times (M/M/2/GD/∞/ ∞)

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- The average number of customers present in the system

$$\boldsymbol{L = L_q + L_s = L_q + \frac{\lambda}{\mu}}$$

Tiger and Wheels – service times (M/M/2/GD/∞/ ∞)

- $\lambda = 2$ [cars/hour] and $\mu = 3$ [cars/hour] with $s = 2$ parallel services.
- The average number of customers present in the system

$$\boldsymbol{L = L_q + L_s = L_q + \frac{\lambda}{\mu} = \frac{P(j \geq s)\rho}{1-\rho} + \frac{\lambda}{\mu}}$$

$$\boldsymbol{P(j \geq s) = \frac{(s\rho)^s \pi_0}{s!(1-\rho)}}$$

$$\boldsymbol{\pi_0 = \left(\sum_{i=0}^{s-1} \frac{(s\rho)^i}{i!} + \frac{(s\rho)^s}{s!(1-\rho)}\right)^{-1}}$$

$$\rho = \lambda/(s\mu)$$

Tiger and Wheels – service times (M/M/2/GD/∞/ ∞)

- $\lambda = 2$ [cars/hour] and $\mu = 3$ [cars/hour] with $s = 2$ parallel services.
- The average number of customers present in the system

$$\boldsymbol{L = L_q + L_s = L_q + \frac{\lambda}{\mu} = \frac{P(j \geq s)\rho}{1-\rho} + \frac{\lambda}{\mu} = \frac{\frac{1}{6}\cdot\frac{1}{3}}{1-\frac{1}{3}} + \frac{2}{3} = \frac{3}{4}} \textbf{ cars}$$